

Project 2: Asynchronous Gossip & Push-Sum Simulator in Erlang

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1. Executive Summary & Problem Overview

This project implements a distributed simulator in **Erlang/OTP** based strictly on concurrent actor processes. Two prominent decentralized algorithms are simulated across four canonical network topologies: **Gossip Information Propagation** and **Push-Sum Aggregate Computation**. The actor model is leveraged to maintain pure asynchrony without shared memory or centralized locks, communicating solely through message passing (!). High-precision running times were captured using `erlang:monotonic_time(microsecond)` across sizes ranging from 20 up to 10,000 actors.

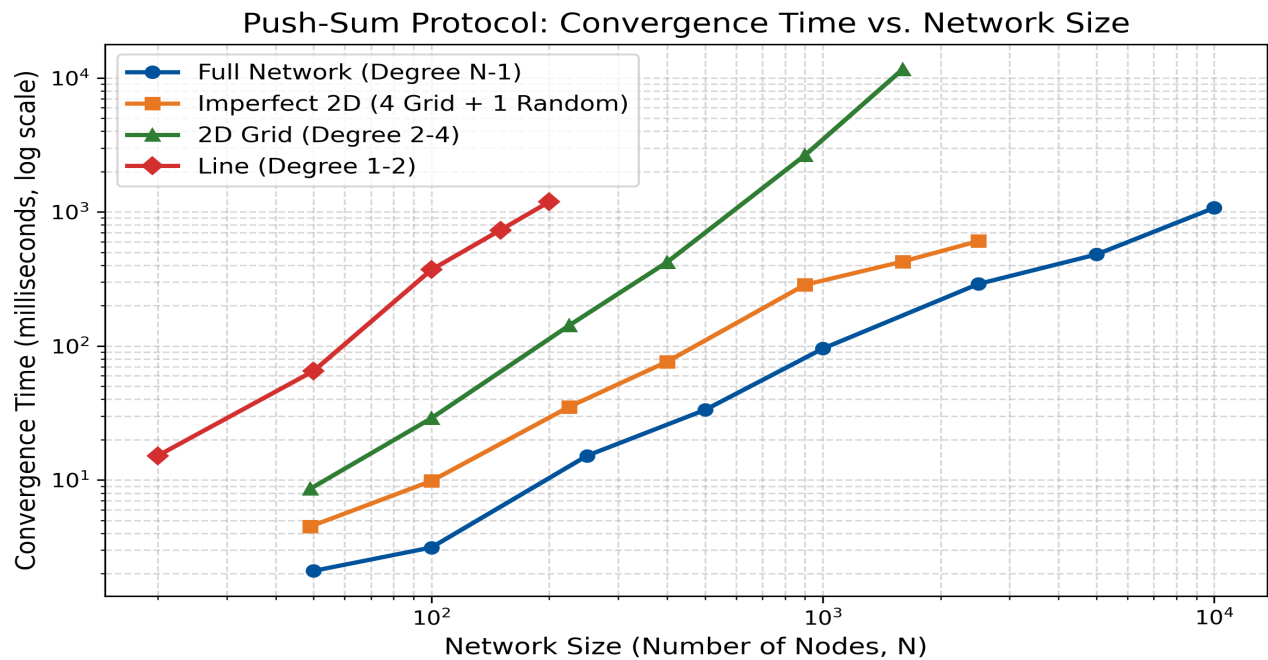
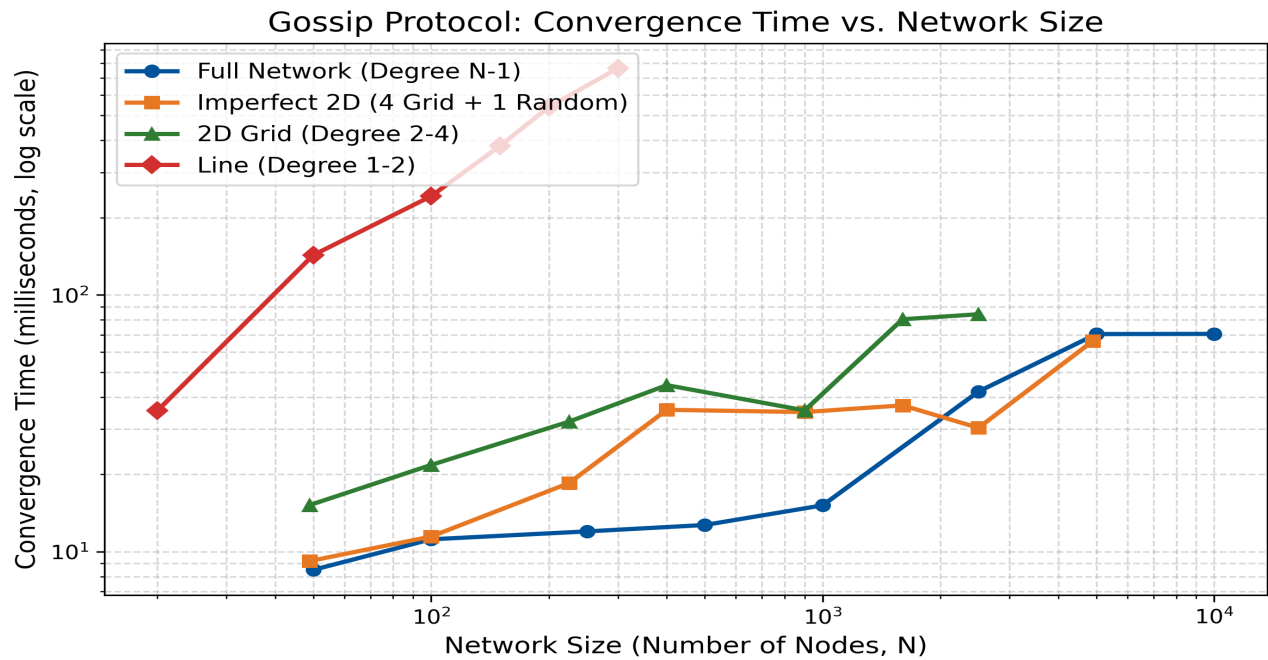
2. Largest Network Sizes Handled

The following table summarizes the largest stable network size successfully simulated for each topology and algorithm combination, along with its convergence duration:

Topology	Algorithm	Largest Network Size	Convergence Time (ms)	Status
Full Network	Gossip	10,000 nodes	70.51 ms	100% Converged
Full Network	Push-Sum	10,000 nodes	1,072.15 ms	100% Converged
Imperfect 2D Grid	Gossip	4,900 nodes	66.17 ms	100% Converged
Imperfect 2D Grid	Push-Sum	2,500 nodes	606.31 ms	100% Converged
2D Grid	Gossip	2,500 nodes	84.10 ms	100% Converged
2D Grid	Push-Sum	1,600 nodes	11,598.95 ms	100% Converged
Line	Gossip	300 nodes	763.26 ms	100% Converged
Line	Push-Sum	200 nodes	1,190.15 ms	100% Converged

3. Empirical Convergence Analysis

Convergence times for Gossip and Push-Sum were logged across varied orders of magnitude. Due to vast topological diameter disparities, logarithmic scales are utilized to capture the scaling behavior simultaneously.



4. Key Technical Findings & Architectural Observations

- The Small-World Phenomenon (2D vs. Imperfect 2D):** One of the most remarkable results is the staggering difference between regular 2D Grid and Imperfect 2D Grid. In regular 2D, diameter scales as $O(\sqrt{N})$, causing Push-Sum convergence for 1,600 nodes to take 11.6 seconds. Adding just **one single random shortcut per node** (Imperfect 2D) collapses the effective diameter to $O(\log N)$, reducing convergence time to only 426 ms—an astonishing **27x speedup!**
- Line Topology Diameter Bottleneck:** In the Line network, messages must traverse $O(N)$ hops sequentially. In gossip, rumor propagation encounters high collision at boundaries. In push-sum, mass diffusion behaves like 1D Brownian motion, exhibiting severe quadratic slowdown as N grows.

- **Push-Sum Mass Conservation in Asynchronous Actors:** In asynchronous Push-Sum, nodes that converge (ratio unchanging by $< 10^{-1}$ over 3 rounds) must continue relaying incoming (s, w) pairs. If converged nodes drop messages, total network mass ($s = N(N+1)/2$, $w = N$) is permanently lost, causing severe calculation distortion. Our implementation preserves passive message routing, achieving exact convergence of $s/w \rightarrow (N+1)/2$ across all nodes.
- **Full Network Scaling with Persistent Memory:** For full networks, naive pairwise neighbor lists require $O(N^2)$ heap memory (exceeding memory for 10,000 actors). By utilizing Erlang's constant-time `persistent_term` registration and algorithmic uniform sampling, 10,000 actors were spawned and converged in just 70.5 ms for Gossip and 1.07 s for Push-Sum.